

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION**Annual Examination 2026****MATHEMATICS — SSC-I (9th Grade)****Total Marks: 65****Time Allowed: 2 Hours 30 Minutes****Version: 2****General Instructions:**

1. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
2. **Section B** (Short Questions): Attempt **11**, choosing main **OR** alternative. Each **3 marks**.
3. **Section C** (Long Questions): Attempt **4**, choosing main **OR** alternative. Each **5 marks**.
4. Use black or blue ink only. Pencil is not allowed except for diagrams and number lines.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)*Encircle the correct option. All questions are compulsory.*

#	Question	A	B	C	D	Answer
1	The set of all elements of U which are not in A is called:	Difference	Union	Intersection	Complement of A	ⒶⒷⒸⒹ
2	If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $A \cap B$ is:	$\{1, 2, 3, 4, 5\}$	$\{3\}$	$\{1, 2\}$	$\emptyset$	ⒶⒷⒸⒹ
3	If $n(A) = 4$ and $n(B) = 3$, then $n(A \times B)$ equals:	7	12	4	3	ⒶⒷⒸⒹ
4	The set of all first elements of ordered pairs in a binary relation is called:	Range	Domain	Set	Function	ⒶⒷⒸⒹ
5	The factorization of $a^2 - b^2$ is:	$(a + b)^2$	$(a - b)^2$	$(a + b)(a - b)$	$a^2 + b^2$	ⒶⒷⒸⒹ
6	The factors of $x^2 + 7x + 12$ are:	$(x+3)$ $(x+4)$	$(x+2)$ $(x+6)$	$(x+1)$ $(x+12)$	$(x-3)$ $(x-4)$	ⒶⒷⒸⒹ
7	The HCF of $12x^2y^3$ and $18xy^2z$ is:	$6x^2y^2z$	$6xy^2$	$36x^2y^3z$	$3xy^2$	ⒶⒷⒸⒹ
8	If the product of HCF and LCM of two polynomials is $x^3 - y^3$, then the product of the polynomials is:	$x^3 - y^3$	$(x - y)^3$	$x^2 - y^2$	$x^2 + y^2$	ⒶⒷⒸⒹ
9	The solution of $2x + 3 = 11$ is:	3	4	5	6	ⒶⒷⒸⒹ
10	$ -6 $ equals:	-6	6	0	12	ⒶⒷⒸⒹ
11	Which of the following is a strict inequality?	$x \geq 3$	$x \leq 5$	$x > 2$	$x = 4$	ⒶⒷⒸⒹ
12	When both sides of an inequality are multiplied by a negative number, its direction:	Stays same	Reverses	Disappears	Doubles	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt **11** questions. Choose either the main question **OR** its alternative.

Q.	Question	Marks	OR	OR Question	Marks
2(i)	Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6, 7\}$. Find: (a) $A \cup B$ (b) $A \cap B$ (c) $A - B$	3	OR	Let $U = \{1, 2, 3, \dots, 10\}$ and $A = \{2, 4, 6, 8\}$. Find: (a) A^c (b) $n(A)$ (c) $n(U)$	3
2(ii)	Write each set in set-builder form: (a) $\{2, 4, 6, 8, 10\}$ (b) $\{1, 4, 9, 16, 25\}$ (c) $\{-3, -2, -1, 0, 1, 2, 3\}$	3	OR	If $A = \{1, 2, 3\}$ and $B = \{3, 4\}$, find $A \times B$ and $B \times A$. Show that $A \times B \neq B \times A$.	3
2(iii)	In a class of 50 students, 30 like Mathematics, 25 like Physics, and 10 like both. How many students like at least one subject? How many like neither?	3	OR	Find the values of x and y if $(2x - 1, y + 3) = (5, 7)$.	3
2(iv)	Let $R = \{(1, 2), (2, 4), (3, 6), (4, 8)\}$. Find: (a) Domain of R (b) Range of R (c) R^{-1}	3	OR	If $A = \{a, b, c\}$, find the number of all possible binary relations in $A \times A$.	3
2(v)	Factorize: (a) $x^2 - 9x + 20$ (b) $4x^2 - 25y^2$	3	OR	Factorize: (a) $x^3 + 27$ (b) $8a^3 - 27b^3$	3
2(vi)	Factorize the trinomial: $3x^2 + 11x + 6$	3	OR	Factorize: $x^4 + x^2y^2 + y^4$	3
2(vii)	Find the HCF of: $x^2 - 4$ and $x^2 + 4x + 4$	3	OR	Find the LCM of the monomials: $6x^2y$, $9xy^2$, $12x^3$	3
2(viii)	Simplify: $(x^2 - 1)/(x^2 + 2x + 1) \times (x + 1)/(x - 1)$	3	OR	Find the square root of $4x^2 + 12xy + 9y^2$ by factorization.	3
2(ix)	Solve for x : $(3x - 1)/4 = (x + 2)/3$	3	OR	Solve the radical equation: $\sqrt{x + 5} = 4$. Check your solution.	3
2(x)	Solve: $ 2x - 3 = 5$	3	OR	Solve the absolute value equation: $ x - 4 = 7$	3
2(xi)	Solve the inequality: $3x - 5 < 2x + 7$, $x \in \mathbb{R}$	3	OR	Solve the compound inequality: $-3 < 2x + 1 \leq 7$, $x \in \mathbb{R}$	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Attempt **4** questions. Choose either the main question **OR** its alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Verify the associative law of union $(A \cup B) \cup C = A \cup (B \cup C)$ for the following sets both analytically and by a Venn diagram: $U = \{1, \dots, 9\}$, $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $C = \{5, 6, 7\}$	5	OR	Verify the distributive law $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ for: $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{5, 6, 7, 8\}$	5
3(ii)	Factorize completely: (a) $x^4 + 4y^4$ (b) $(x^2 - 5x + 2)(x^2 - 5x + 4) - 3$	5	OR	Find HCF and LCM of $x^3 + 2x^2 - 3x$ and $x^3 + x^2 - 6x$. Verify that $\text{HCF} \times \text{LCM} = \text{product of the polynomials}$.	5
3(iii)	Find the square root of $x^4 - 6x^3 + 13x^2 - 12x + 4$ by the division method and state whether it is a perfect square.	5	OR	Simplify: $((x^2 - 4)/(x^2 - 9)) \div ((x + 2)/(x + 3)) \times ((x^2 - 6x + 9)/(x^2 + 2x))$	5
3(iv)	Solve the compound inequality $5 < 2x + 3 \leq 17$, $x \in \mathbb{R}$. Show the solution set on a number line.	5	OR	The sum of three consecutive integers is greater than 30 and less than or equal to 72. Find all possible values of the smallest integer.	5

Invigilator's
Signature

Candidate's
Signature

Chapters Covered: Unit 3 — Sets and Relations | Unit 4 — Factorization | Unit 5 — Linear Equations & Inequalities
— End of Question Paper —